

Title: Text Embeddings and Cosine Similarity

Introduction:

Text embeddings are numerical representations of text that capture the semantic meaning of sentences. They convert sentences into vectors so that computers can compare their meanings mathematically.

Importance of Embeddings in Semantic Search:

Embeddings help search systems understand the meaning of a query rather than only matching exact keywords. Similar sentences are represented by vectors that are closer to each other. Therefore, embeddings are useful for finding relevant information from large collections of documents.

Importance in LLM Applications:

Embeddings are widely used in Large Language Model applications such as semantic search, recommendation systems, document retrieval, question answering and Retrieval-Augmented Generation (RAG). They help applications identify relationships between different pieces of text.

Cosine Similarity:

Cosine similarity measures the similarity between two sentence embedding vectors. A higher score indicates that the sentences have more similar meanings, while a lower score indicates less semantic similarity.

Conclusion:

In this activity, 20 sentences were converted into numerical embeddings using a pre-trained Sentence Transformer model. Ten sentence pairs were compared using cosine similarity. The results showed that sentences with similar meanings generally received higher similarity scores. This demonstrates how embeddings can be used to understand and compare the semantic meaning of text.